

Numerical Analysis Exercise 2

2. Use the bisection method to find p_3 for $f(x) = \sqrt{x} - \cos x = 0$ in $[0,1]$.

```
1 %%2.2寻找p3
2 clear all
3 clc
4 syms x;
5 n=0;%迭代次数
6 a=0;b=1; x0=0; p=x0;
7 while (n~=3)
8     if(f(p)==0)
9         x0=p;
10        break;
11    end
12    if(f(a)*f(p)<0)
13        b=p;
14        p=(a+b)/2;
15    else
16        a=p;
17        p=(a+b)/2;
18    end
19    x0=p;
20    n=n+1;
21 end
22 p
23 n
24 function f=f(x)
25 f=x^(1/2)-cos(x);
26 end
```

```
1 p = 0.6250
2 n = 3
```

3. Use the bisection method to find solutions accurate to 10^{-4} for the following problem:

(1) $x^3 - 2^{-x} = 0$, for $0 \leq x \leq 1$;

```
1 %%定义的函数
2 function z=f_Bis(f,a,b,accurate)
3 x0=0;
4 p=x0;
5 n=0;
6 z=0;
7 persistent x;
8 while (abs(f(p)-0)>accurate)
9     if(f(p)==0)
10         x0=p;
11         break;
12     end
13     if(f(a)*f(p)<0)
14         b=p;
15         p=(a+b)/2;
16     else
17         a=p;
18         p=(a+b)/2;
19     end
20     x0=p;
21     n=n+1;
22 end
23 x0
24 n
25 end
```

```
1 %%调用函数
2 y = @f
3 z = f_Bis(y,0,1,1e-4);
4 function y = f(x)
5 y=x^3-2^(-x);
6 end
```

```
1 x0 = 0.8262
2 n = 14
```

4. Find an approximation to $\sqrt{3}$ correct to within 10^{-4} using the bisection method.

[Hint: Consider $f(x) = x^2 - 3$]

```
1 %%调用函数
2 y = @f
3 z = f_Bis(y,0,2,1e-4);
4 function y = f(x)
5 y=x^2-3;
6 end
```

```
1 x0 = 1.7321
2 n = 14
```